Program : **Diploma in Instrumentation Engineering**	
Course Code : **6081C**	Course Title: **Aircraft Instruments**
Semester : **6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To introduce Instrumentations and sensors used in Aircrafts
- To learn the basics of Flight data recording

Course Prerequisites:

Topic	Course code	Course Title	Semester
Sensors & Instruments used in Aircrafts.		Sensors & Transducers	3
Variable measurements techniques.		Process Variable Measurements	4

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate basic Requirements of Air Craft Instruments	14	Understanding
CO2	Explain Pitot static flight Instruments	15	Understanding
CO3	Outline principles of Gyro Instruments	15	Understanding
CO4	Illustrate different sensors used in air crafts and flight data recording	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Illustrate basic Requirements of Air Craft Instruments**		
M1.01	Illustrate Flight and Navigation Instruments	3	Understanding
M1.02	Explain Anatomy of airplane with emphasis on control surfaces	3	Understanding
M1.03	Illustrate History of Aircrafts	2	Understanding
M1.04	Illustrate the basics of aerodynamics	2	Understanding
M1.05	Outline Types of Displays in Air Crafts	4	Understanding
Contents: Flight and Navigation Instruments - Anatomy of airplane and space vehicle with emphasis on control surfaces - History of Aircrafts - Types of Displays in Air Craft - Quantitative displays - Circular Scale-Linear and Non Linear scale displays - High range long scale displays - Straight scale display - Digital display-Dual Indicator - Coloured Display - Direct display - Head up display - LED and LCD Display.			
CO2	**Explain Pitot static flight Instruments**		
M2.01	Outline Pitot static flight Instruments	4	Understanding
M2.02	Explain Sensing and transmission of Pitot pressure and Static Pressure	3	Understanding
M2.03	Explain Aneroid Barometer and altimeter.	4	Understanding
M2.04	Illustrate Air speed Indicator	4	Understanding

	Series Test – I	1	
Contents: Pitot static flight Instruments - Pitot Pressure - sensing and transmission of Pitot pressure and Static Pressure - Pitot static Probe - heating circuit arrangement in pitot tube - Aneroid Barometer and altimeter. Air speed Indicator - Mach Number - Mach meter - Vertical Speed Indicator			
CO3	**Outline principles of Gyro Instruments**		
M3.01	Outline Gyro Instruments	2	Understanding
M3.02	Explain pneumatic and electric method of driving Gyroscope rotor	4	Understanding
M3.03	Illustrate Gyro Horizon	3	Understanding
M3.04	Illustrate Electrically operated Engine speed Indicator	3	Understanding
M3.05	Explain Tacho probe and Indicator System.	3	Understanding
Contents: Gyro Instruments - altitude Indication - Pitch, Bank and Turn - Gyroscope - degrees of freedom of Gyroscope - pneumatic and electric method of driving Gyroscope rotor - Gyro Horizon - principle of gyrohorizon - Electrically operated Engine speed Indicator - Tacho probe and Indicator System.			
CO4	**Illustrate different sensors used in air crafts and flight data recording**		
M4.01	Outline Type of Temperature sensors used in air crafts	1	Understanding
M4.02	Illustrate Thermocouples	2	Understanding
M4.03	Illustrate Radiation Pyrometers	2	Understanding
M4.04	Explain Pressure gauges and pressure transmitters	3	Understanding
M4.05	Explain Capacitance type Fuel gauge system	2	Understanding
M4.06	Illustrate flight data recording	3	Understanding
M4.07	Explain Accelerometers	2	Understanding
	Series Test – 2	1	
Contents: Temperature sensors used in air - surface contact type and immersion type thermocouples - Radiation Pyrometer for exhaust gas temperature measurement- direct reading Pressure gauges- Inductor Pressure Transmitter- Pressure switches- fuel quantity measurement methods in air craft- Capacitance type Fuel gauge system used in air crafts.-flight data			

recording - the mandatory parameters recorded - Accelerometer - trace recording and electromagnetic recording

Text /Reference:

T/R	Book Title/Author
R1	E H J Pallett -Aircraft Instruments–Principles and Applications- Pitman Publishing Inc
R2	Ernest O Doebelin, Dhanesh N Manik , Measurement Systems-Application and Design,5th Edition, Tata McGraw Hill, 2007
R3	Max.F.Henderson - Aircraft Instruments and Avionics-.Publisher Jeppesen
R4	C.A.Wlliams -Aircraft Instruments-Sterling Book House
R5	Jewel B Barlow, William H. Rae, Jr. , Alan Pope , Low-Speed Wind Tunnel Testing, John Wiley, Third Edition, 1999

Online resources:

Sl. No	Web site link
1	https://www.skybrary.aero/index.php/Aerofoil
2	http://aviationknowledge.wikidot.com/aviation:aerofoil
3	https://www.britannica.com/technology/history-of-flight
4	https://www.decodedscience.org/slats-slots-and-spoilers-lift-modifying-devices-onairplane-wings/12105